

EX:9 Create a Simple Android Application to Convert Text to Speech Using the TextToSpeech API

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center">
    <EditText
        android:id="@+id/editText"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Text"
        android:inputType="text" />
    <Button
        android:id="@+id/btnSpeak"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Speak"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Java Code (MainActivity.java)

```
package com.example.texttospeechapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.speech.tts.TextToSpeech;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import java.util.Locale;

public class MainActivity extends AppCompatActivity {

    EditText editText;

    Button btnSpeak;

    TextToSpeech textToSpeech;

    @Override

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        editText = findViewById(R.id.editText);
        btnSpeak = findViewById(R.id.btnSpeak);
        textToSpeech = new TextToSpeech(this, status -> {
            if (status != TextToSpeech.ERROR) {
                textToSpeech.setLanguage(Locale.US);
            }
        });

        btnSpeak.setOnClickListener(new View.OnClickListener() {

            @Override
```

```
public void onClick(View v)
{
    String text = editText.getText().toString();
    textToSpeech.speak(text, TextToSpeech.QUEUE_FLUSH, null, null);
}
});
}

@Override
protected void onDestroy() {
    if (textToSpeech != null) {
        textToSpeech.stop();
        textToSpeech.shutdown();
    }
    super.onDestroy();
}
}
```

OUTPUT

